

HDRNet: High-Dimensional Regression Network for Point Cloud Registration

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Abstract

Abstract-3D point cloud registration is a crucial topic in the reverse engineering, computer vision and robotics fields. The core of this problem is to estimate a transformation matrix for aligning the source point cloud with a target point cloud. Several learning-based methods have achieved a high performance. However, they are challenged with both partial overlap point clouds and multiscale point clouds, since they use the singular value decomposition (SVD) to find the rotation matrix without fully considering the scale information. Furthermore, previous networks cannot effectively handle the point clouds having large initial rotation angles, which is a common practical case. To address these problems, this paper presents a learning-based point cloud registration network, namely HDRNet, which consists of four stages: local feature extraction, correspondence matrix estimation, feature embedding and fusion and parametric regression. HDRNet is robust to noise and large rotation angles, and can effectively handle the partial overlap and multi-scale point clouds registration. The proposed model is trained on the ModelNet40 dataset, and compared with ICP, SICIP, FGR and recent learning-based methods (PCRNet, IDAM, RGMNet and GMCNet) under several settings, including its performance on moving to invisible objects, with higher success rates. To verify the effectiveness and generality of our model, we also further tested our model on the Stanford 3D scanning repository.

Keywords: point-based methods, methods and applications, 3D shape matching, modelling

CCS Concepts: • Computing methodologies → Matching

1. Introduction

With the continuous development of diverse scanning equipment, 3D point clouds have become a popular format of 3D data, due to their adaptiveness and descriptiveness. Therefore, the point cloud registration becomes a fundamental and crucial topic in several computer graphics and computer vision applications, such as 3D reconstruction [YGL*19, CLHC14], 3D localization [CBCB18, CLF*20] and pose estimation [PCS15, WWLW21], for example. The point cloud registration consists in computing a rigid change to align the scanned source point cloud and the target point cloud. Therefore, several parts of point clouds can be merged to further form a complete point cloud.

The registration of point clouds has been studied for several decades. The most widely used traditional point cloud registration method is the Iterative Closest Point (ICP) [BM92, SHT09]. This method has the drawback of local optimum. A series of ICP variants have been proposed to overcome this drawback, such as fast

global registration (FGR) [ZPK16], GO-ICP [YLJ13] and Scale Iterative Closest Point (SICP) [ZXH*17]. However, the computational speed of these methods is very slow, compared with that of ICP. Another traditional strategy for registering point clouds, such as RANSAC [RBB09, STD14], consists in first extracting the corresponding point pairs, and then obtaining the rigid changes from the point pairs. However, this approach usually can only complete the coarse registration.

Several recent methods used deep networks to establish corresponding relationships without transformation initialization, such as DCP [WS19a], RPM-Net [YL20] and IDAM [LZX*19], for example. These methods highly improve the accuracy and the computation speed, and overcome the local optimum issue. However, several challenging issues that should be addressed, still exist. More precisely, three types of registration problems commonly exist.

First, two partial overlapping point clouds usually exhibit large scale of outliers, which refer to the points without corresponding

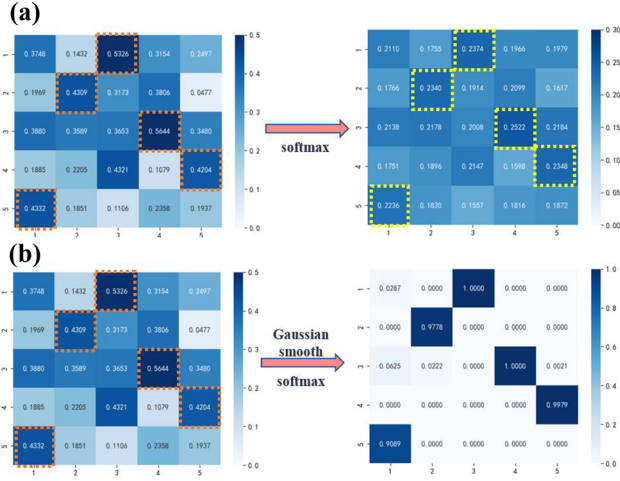


Figure 1: Comparison diagram of corresponding matrix without Gaussian smoothing and Gaussian smoothing. (a) Column-wise softmax for similar matrices. Data marked in yellow dotted rectangle represent the element with the largest numerical value in a column. It can be seen that the data distribution in the matrix is relatively uniform and has no significant information. (b) Column-wise Gaussian smoothing and softmax for similar matrices. Orange represents the element with the largest numerical value in a column. It can be seen that the smoothed matrix is close to the binary correspondence matrix.

relationship in the target point cloud [FLLW21]. Consequently, the registration problem will be more difficult to solve. Second, many point cloud acquisition mechanisms have inclined to matured, such as Kinect and Lidar, for example. However, each sensor has specific physical measures, indicating that point clouds may be captured at multi-scales. Thus, the registration of point clouds with different scales is also a necessary and challenging research area [PWF*14, HZF*17]. Although SICP [ZXH*17] can deal with this case, its main limitation is the sensitiveness to the initial transformation. Third, large initial

rotation angles between source and target point clouds, which is a common case in practice, has considerably lowered the performance of the traditional ICP-based methods and the recent deep learning-based methods. To tackle the problems of mentioned above, this paper proposes a deep network for point cloud registration, namely the high-dimensional regression network (HDRNet).

Motivation. First, the previous methods usually only employ the softmax function to compute the corresponding matrix, however, as shown in Figure 1a, the data distribution in the resulted corresponding matrix is relatively uniform and contains few significant information. Therefore, we first apply the Gaussian function on the corresponding matrix to filter out the significant information, and the results of which are shown in Figure 1b. Furthermore, both the point cloud and its local features are disordered, we thus apply the corresponding matrix to the local features, consequently, the local features of the two point clouds can be basically ordered, as shown in Figure 2. Hence, the initial rotation angle and the scale factor will not affect the results. Third, when there is only a partial overlap be-

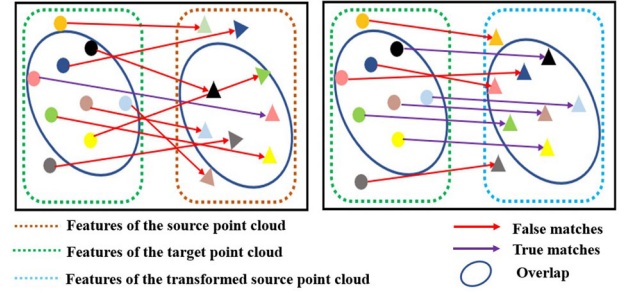


Figure 2: A toy example used to represent the function of the corresponding matrix.

tween the source point cloud and the target point cloud, using the distance between the predicted target point cloud and the target point cloud as a loss will not learn the ground truth very well. Therefore we use the distance between the point cloud that is generated by applying the ground truth to the source point cloud and the predicted target point cloud as loss, which can effectively solve the partially overlap case.

Based on the above motivation, the proposed HDRNet consists of four parts: (a) a feature extraction layer that maps a three dimensional point cloud to a higher dimension, which is equivalent to

calculating the local features of each point, (b) the correspondence matrix estimation layer, which aims to learn a correspondence matrix that explicitly represents the correspondence between two local features, (c) the feature embedding and fusion layer. It consists in computing the inner product matrix of the local features of two point clouds, instead of directly using maximum pooling to extract the global features of two point clouds. This strategy fully improves the availability of local point cloud features, and can effectively handle point cloud registration with large initial angles. 2DCNN is then used to further extract the features and expand them into one-dimensional tensors. (d) The parametric regression layer, where the full connection layer is used to regress pose information and introduce a scale factor at the end. This can effectively handle point cloud registration with multi-scales. As for the network training, the EMD loss function used in PCRNet [SLG*19] is improved. This can effectively resolve point cloud registration with partial overlap. HDRNet is tested on the modelnet40 dataset and on the Stanford 3D scanning Repository. The obtained qualitative and quantitative results demonstrate that HDRNet outperforms the typical traditional and deep learning-based methods. A simple result is shown in Figure 3.

The main contributions of this paper are summarized as follows:

- A highly accurate point cloud registration network structure is presented to handle the registration of point clouds with multiple scales, large initial rotation angles and partially overlapping cases.
- We apply the Gaussian function on the corresponding matrix, and design a novel feature embedding and fusion module to extract global features, which reduce the impact of large initial rotation angles and scale factors.

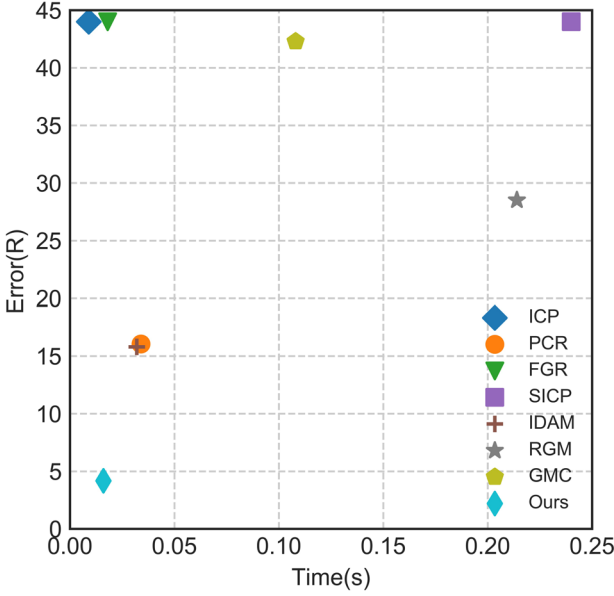


Figure 3: Error (MIE(R))—time(s) comparison of different methods. Our HDRNet achieves the lowest error (R) at approximately the fastest speed. The experiment was evaluated in a multi-scale environment. The time range is limited to 0–0.25 s, the error range is limited to 0–45.

- We introduce a new Earth Mover’s Distance (EMD) as a loss function that minimizes the distance between the predicted target point cloud and the corresponding target point cloud. Thus, in the process of reverse propagation, outliers can be effectively identified and given a smaller weight. Consequently, the registration of point clouds with partial overlapping parts can be realized.
- The proposed method is compared with several methods of the literature on partial–partial overlap, noisy, multi-scale and invisible datasets. The qualitative and quantitative experiment results verify the feasibility of the proposed method.

2. Related Work

2.1. Non-learning-based method

ICP [BM92, SHT09] is the most classical local point cloud registration algorithm. It first performs initial changes to find the nearest neighbours in the target point cloud, then calculates the rigid transformation using the SVD method, and finally iterates repeatedly until the optimal solution is reached. Although ICP achieves a high accuracy, it depends on the precision of the initial position, often falling into local optimum due to large initial disturbances. To solve these shortcomings of ICP, several variants [SHT09, ZXH*17, Rus19, RL01, CSSK02] have been suggested. For instance, trimmed ICP [CSSK02] increases the ability of ICP to handle partially overlapping point clouds. SICP [ZXH*17] increases the ability of ICP to handle point clouds of multi-scales. Sparse ICP [BTP13] can effectively solve the registration problem of sparse point clouds. Simultaneously, several methods [JH02, RXZC17, ZL18] have been offered to improve the calculation efficiency. However, these methods do not address the sensitivity to initial disturbances.

Global registration is used to solve the problem of point cloud registration with large rotation and shift [LJQ17]. A set of branch and bound-based (BnB) methods [YLJ13, CPK*19a, LCSW18, CP16] has been put forward to solve the problem of local registration initial conditions, such as go-icp [YLJ13], which uses branch definition to calculate rigid changes, with a high time complexity reaching $O(N^2)$. RANSAC [RBB09, STD14] is another effective global registration method using usually the corresponding points on the point clouds as the input. Other methods use other strategies to optimize the objective function, such as FGR [ZPK16], which uses FPFH descriptors and alternating optimization schemes for point cloud registration, achieving the highest performance. However, FGR is sensitive to noise and cannot effectively handle the part-to-part point clouds registration.

2.2. Deep learning-based method

RANSAC-based: Several methods [EAF17, CGKW18, EABMD18, LZL*18, CLX*19, YWTS20, SSS19, CPK19b, BLZ*20, LZP*20, AHY*21, ZSN*17] use the deep learning to make local feature descriptors of the point clouds. For instance, FCGF [CPK19b] uses a UNet architecture with skip connections and residual blocks to extract fully convolutional features. D3Feat [BLZ*20] uses KPConv as a backbone to predict feature and point saliency. 3D Match [ZSN*17] first voxelized point clouds and then used 3D CNN to extract features descriptors. The extracted feature descriptors are finally fed to the RANSAC channel, in order to complete the point clouds registration. CentroidReg [ZLWY21] uses the proposed RegistrationNet to extract hybrid rotation-invariant features (HRIF) combining global information of the entire 3Dshape and local structures in the local area to describe each point, and use the Reliable Correspondence Generating (RCG) module to generate correspondences in the overlapping area. This greatly improves the accuracy of the model. GMCNet [PCL21] uses handcrafted features and Transformation-robust Point Transformer (TPT) module generate local features, and uses a Hierarchical Graphic Modelling (HGM) module to encode robust descriptors. The above RANSAC-based method increases the performance at the expense of the computational time. Furthermore, mistakenly removal of non-overlapping points from the target point cloud may produce false matches.

Global feature-based methods: With the advent of PointNet [CSKG17], and DGCNN [WSL*18], a series of end-to-end point cloud registration networks [YPL*19, WCLF19, GFFdQ20, HMZ20] have been proposed. For instance, PointnetLK [AGSL19] first uses PointNet [CSKG17] to extract the global features of point clouds, then uses Lucas and Kanade (LK) algorithm to minimize the distance between the two global features and finally iterates repeatedly to achieve the optimal solution. PCRNet [SLG*19], uses the deep neural network instead of Lucas and Kanade (LK) algorithm, in order to increase the applicability of the model. They capture global features for full-to-full registration, resulting in massive information loss and inaccurate translations. OMNet [XLW*21] learns an overlapping mask to reject non-overlapping regions, which converts part-to-part registrations to same-shape registrations.

Correspondence matching-based methods: The Deep Nearest Point (DCP) [WS19a] calculates the rigid transformation by

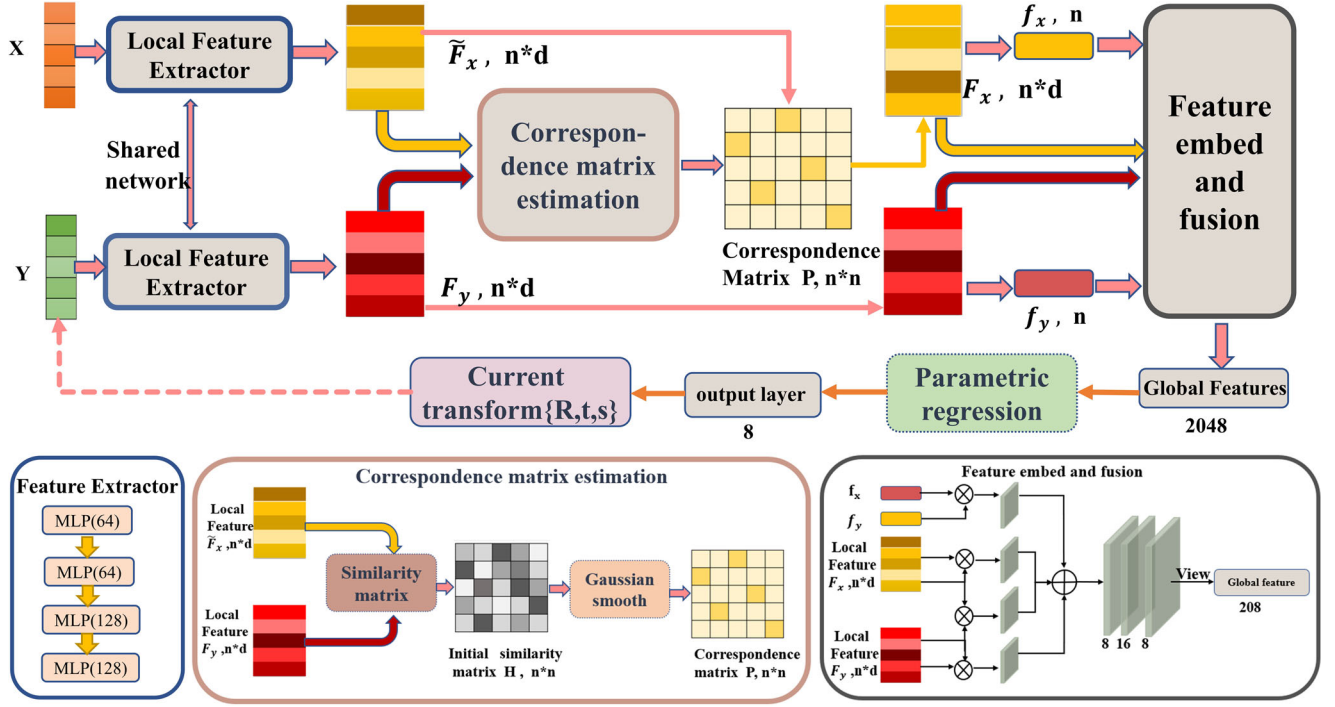


Figure 4: HDRNet architecture. The proposed architecture consists of four main parts: (i) local feature extraction, (ii) correspondence matrix estimation, (iii) feature embedding and fusion, (iv) parameter regression. It takes source and target point clouds as input, applies a multi-layer perceptron (MLP) to extract features, and shares weights among the MLPs. Then, the correspondence matrix estimation between the features is calculated and applied to the features of the source point cloud to obtain the source point cloud features that approximately correspond to the target point cloud features. Finally, the features of the point cloud are embedded with the covariance matrix, and the features are further fused with 2DCNN. The output is a tensor of dimension 8, then the transformation matrix between the source and target point clouds is computed. $\oplus$ represents matrix connection and $\otimes$ represents matrix multiplication.

calculating the soft matching between point clouds, and using the SVD. DeepGMR [YEK*20] solves the point correspondence step by matching points using a Gaussian mixture model whose parameters are estimated by a neural network. IDAM [LZX*19] uses an iterative distance-aware similarity matrix convolution module, which overcomes the disadvantage of inner product matching of eigenvectors, and enables efficient partially to partially point cloud registration. RPMNet [YL20] uses a predictive annealing network that can handle outliers and partially to partially point clouds using Sinkhorn normalization. RGMNet [FLLW21] converts point clouds into graphics and uses a module based on depth map matching to calculate the soft correspondence matrix. The above correspondence-based methods solves the partial-to-partial registration problem indirectly. DeepPRO [LHF*21] uses the Conditional Point Cloud Generation model to directly predict the point-by-point location of the alignment point cloud and to predict the rotation matrix from the quaternion.

3. Problem Description

This section summarizes the problem setting. We design an end-to-end learning framework for point cloud registration, meanwhile taking unordered and unoriented point clouds as input. For a given

source point cloud $X \in \mathbb{R}^{N \times 3}$ and a target point cloud $Y \in \mathbb{R}^{M \times 3}$, N and M represent the number of points in X and Y , respectively. In the experiment, N and M are made equal using a sampling process. Let X_i represents the i th point of the source point cloud and Y_j represents the j th point of the target point cloud, where each point is a vector of its (x, y, z) coordinates. In this work, the 3D point cloud registration aims at estimating the similarity transformation T , including the rotation matrix $R \in \mathbb{R}^{3 \times 3}$, the translation vector $t \in \mathbb{R}^3$ and the scale factor $s \in \mathbb{R}^1$, that aligns X and Y .

4. Method

In this section, the proposed registration network HDRNet is introduced in detail. The overall architecture of the network is presented in Figure 4, which consists of four main parts: (i) local feature extraction, (ii) correspondence matrix estimation, (iii) feature embedding and fusion, (iv) parametric regression.

4.1. Local feature extraction

The input of the proposed network includes both the source point cloud and the target point cloud. The first step is to extract local

feature descriptors from the given two-point clouds. Three key factors need to be overcome in extracting local features: (i) the disorder of the point clouds, (ii) the scale invariance and (iii) the rotation invariance. PointNet [CSKG17] is the first deep network that directly extracts features from the original unstructured point cloud, and that overcomes these three key factors. Thus, we apply MLPs on each point in the point cloud, and in the loss function, we introduce a scale factor to supervise the learning, so that a smaller weight is assigned to a larger scale point cloud. The combined action of the two can guarantee the robustness to scales. Finally, although mini-PointNet [CSKG17] is not robust to scales, mini PointNet can also be utilized to extract local features, meanwhile reducing the amount of parameters.

However, instead of using the global features filtered by max-pooling in mini-Pointnet, we produce an $N \times 128$ tensor as the output, therefore, the local features can be considered and thus forming the local feature descriptor of the source point cloud. More precisely, we put the source cloud which is expressed as an $N \times 3$ tensor, into the local feature extraction layer, as shown in Figure 4. The local feature extraction layer consists of three multi-layered perceptrons of size 64, 64, 128, respectively. The target point cloud uses the same structure as the source point cloud. The local feature extraction layer encodes point clouds into a 128-dimension vector which is expected to have certain semantics. Consequently, the local feature descriptors of the source and the target point clouds are stored in vectors $\tilde{F}(x) = \{\tilde{X}_1^L, \tilde{X}_2^L, \tilde{X}_3^L, \dots, \tilde{X}_i^L, \dots, \tilde{X}_n^L\}$ and $F(y) = \{Y_1^L, Y_2^L, Y_3^L, \dots, Y_j^L, \dots, Y_n^L\}$.

4.2. Correspondence matrix estimation

Our correspondence matrix estimation module aims to learn a correspondence matrix $P \in \mathbb{R}^{N \times N}$ that explicitly represents the correspondence between any two high-dimensional features of $\tilde{F}(x)$ and $F(y)$. Ideally, P should be a binary orthogonal permutation matrix, however, this matrix is non-differentiable and difficult to optimize in neural networks. Therefore, we first compute a similarity matrix with only one prominent element in each column. Then we Gaussian smooth the similarity matrix, quantizing it into a near binary correspondence matrix P . First, we compute the similarity matrix H based on the inverse distance between $\tilde{F}(x)$ and $F(y)$:

$$H_{ij} = e^{-\|\tilde{X}_i^L - Y_j^L\|_2} \quad (1)$$

where $H_{ij} \in H \in \mathbb{R}^{N \times N}$, $\tilde{X}_i^L$ and Y_j^L are the i th and j th point features corresponding to X and Y , respectively. However, H still deviates greatly from the actual corresponding matrix. Assume that H_{ij} obeys a series of Gaussian distributions:

$$H_{ij} \sim N(\mu_j, \sigma_j^2) \quad (2)$$

where μ_j and σ_j^2 are the mean and standard deviation of the j th column of H . To further enhance H , for each element H_{ij} of the j column of H , we use its probability density function to compute the enhanced matrix $\tilde{P}$:

$$\tilde{P}_{ij}^1 = \frac{1}{\sqrt{2\pi}\sigma_j} \exp\left(-\frac{(H_{ij} - M_j)^2}{2\sigma_j^2}\right) \quad (3)$$

where M_j and σ_j^2 are the maximum value and standard deviation of the j th column of H . As shown in Figure 5, Gaussian smoothing increases the distance between the maximum value of each column in the square matrix and the other values. To further expand this distance, we use $\tilde{P}_{ij}^1 = \frac{\tilde{P}_{ij}^1}{m_j}$. m_j is the minimum value of the j th column in matrix $\tilde{P}_{ij}^1$.

Ideally $\sum_{j=1}^n \tilde{P}_{ij}^1 = 1$. However, it is different actually. So, we apply a softmax operation on $\tilde{P}$ to get the corresponding matrix P , that is

$$P_{ij} = \frac{e^{\tilde{P}_{ij}}}{\sum_{j=1}^n e^{\tilde{P}_{ij}}} \quad (4)$$

Since the differences between the maximum value of each column in the matrix and the other values are significantly expended, as shown in Figure 5a,b, a near-binary correspondence matrix can be achieved by applying the softmax, as shown in Figure 5c. Our network can well learn the correspondence matrix, which is illustrated in Figure 6. It can be seen that the data in the matrix has prominent features and important corresponding information. Finally, we permute the $F(x)$ by the learned P approximate to a permutation matrix:

$$F(x) = \tilde{F}(x) \otimes P \quad (5)$$

4.3. Feature embedding and fusion

Using matrices empirically gives better results than using feature vectors, referencing the ablation results. Given vectors $F(x) = \{X_1^L, X_2^L, X_3^L, \dots, X_i^L, \dots, X_n^L\}$ and $F(y) = \{Y_1^L, Y_2^L, Y_3^L, \dots, Y_j^L, \dots, Y_n^L\}$ that are the local feature descriptors of the source and the target point clouds. In particular, X_i^L and Y_j^L are the descriptors corresponding to X_i and Y_j , respectively, Y_j is the nearest neighbour searched in the Euclidean space, based on the L2 distance by applying the ground truth on X_i as

$$Y_j = X_i \otimes T \quad (6)$$

where $\otimes$ and T , respectively, denote matrix multiplication and ground truth. T includes the rotation matrix $R \in \mathbb{R}^{3 \times 3}$, the translation vector $t \in \mathbb{R}^3$ and the scale factor $s \in \mathbb{R}^1$. Because $F(x)$ and $F(y)$ are obtained by the same module and parameters, there is a mapping T^1 between $F(x)$ and $F(y)$, and there is also a definite relationship between T and T^1 :

$$F(y) = F(x) \otimes T^1 \quad (7)$$

where T^1 includes the rotation matrix $R^1 \in \mathbb{R}^{128 \times 128}$, the translation vector $t^1 \in \mathbb{R}^{128}$ and the scale factor $s^1 \in \mathbb{R}^1$.

Since it is impossible to calculate T^1 directly from $F(x)$ and $F(y)$, calculating T^1 by constructing a square matrix of the same size as T^1 is considered an effective strategy. To simplify the computation of the T^1 , we multiply the transpose $F(x)^T$ of vector $F(x)$, on both sides of Equation (7):

$$M(x, y) = M(x, x) \otimes T^1 \quad (8)$$

$$M(x, y) = F(x)^T \otimes F(y) \quad (9)$$

$$M(x, x) = F(x)^T \otimes F(x) \quad (10)$$

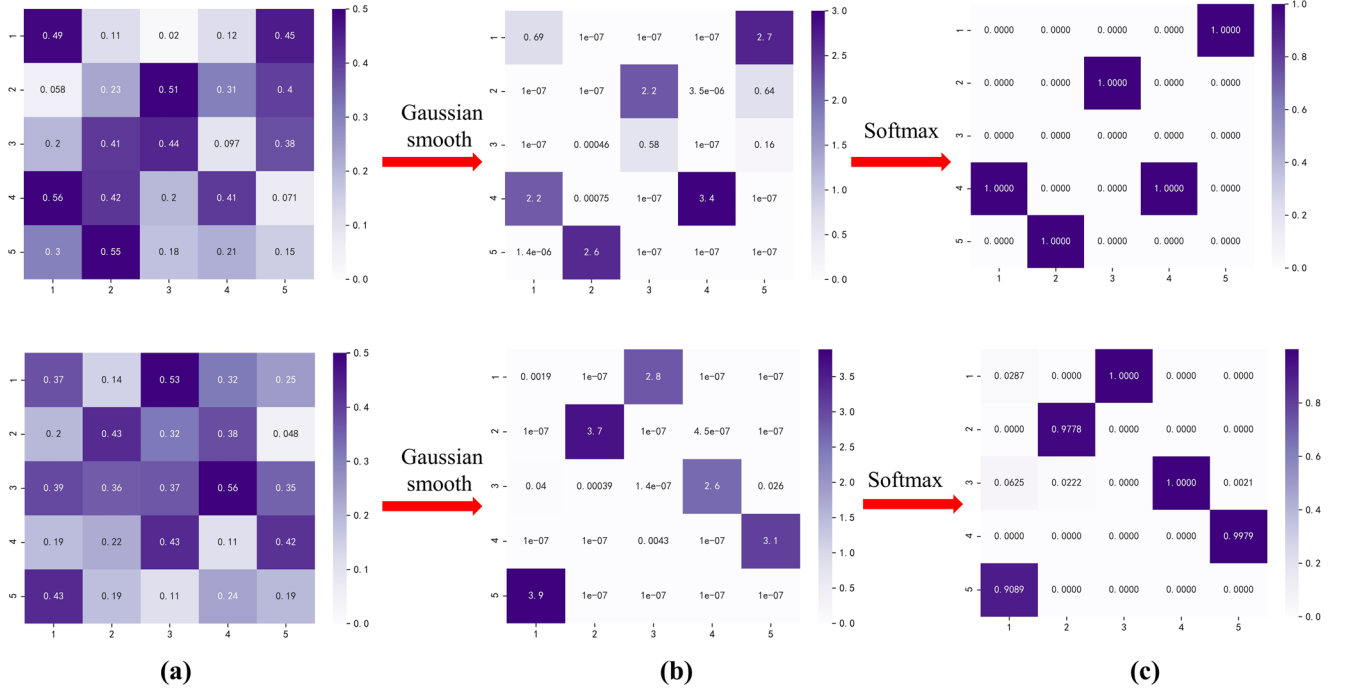


Figure 5: Two examples of Gaussian column-wise Smoothing + softmax to generate a near-binary correspondence matrix. (a) Initial similarity matrix (b) Similarity matrix after Gaussian smoothing (c) Similarity matrix after softmax.

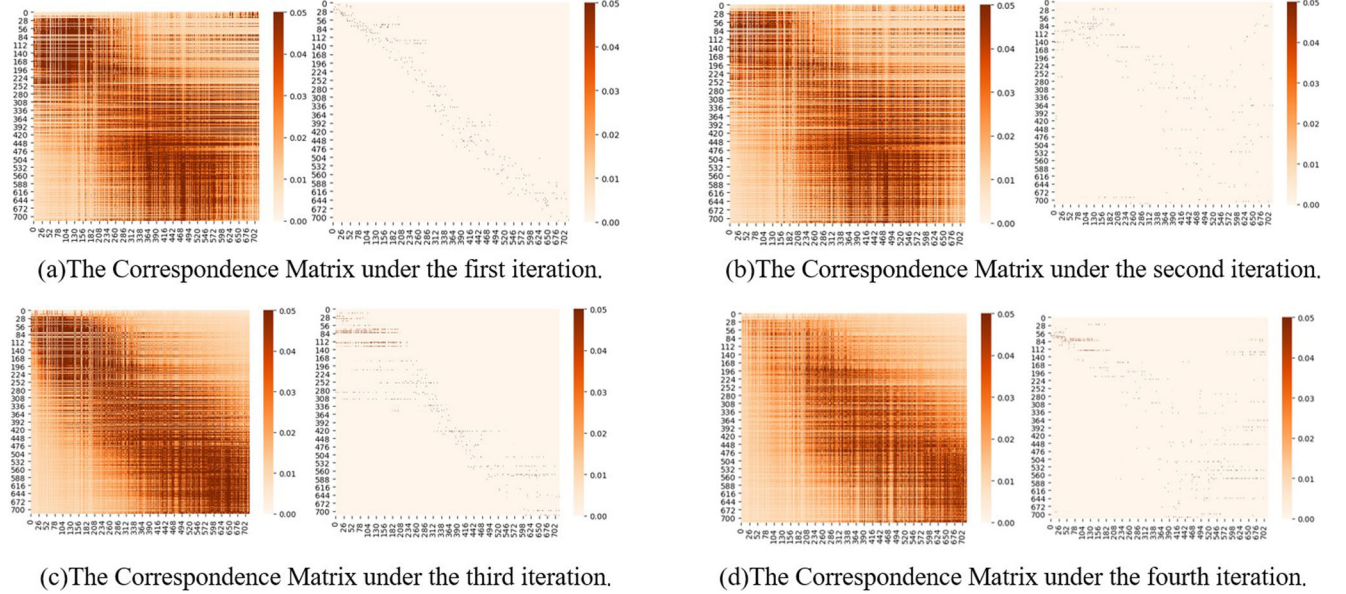


Figure 6: Visualization of the feature similarity matrix and the feature correspondence matrix at different registration stages. (a) The correspondence matrix under the first iteration. (b) The correspondence matrix under the second iteration. (c) The correspondence matrix under the third iteration. (d) The correspondence matrix under the fourth iteration.

where $M(x, y)$ and $M(x, x)$ are all 128×128 matrices. $M(x, y)$ is the inner product result of $F(X)$ and $F(Y)$, $M(x, x)$ is the inner product result of $F(x)$. From Equation (8), T^1 is equal to the inverse matrix of $M(x, x)$ multiplied by matrix $M(x, y)$.

Similarly, T^1 can be obtained from the inverse matrix of $M(x, y)$ and matrix $M(y, y)$, by multiplying the transpose $F(y)^T$ of vector $F(y)$ on both sides of Equation (7):

$$M(y, y) = M(x, y)^T \otimes T^1 \quad (11)$$

$$M(y, y) = F(y)^T \otimes F(y) \quad (12)$$

where $M(y, y)$ is the inner product matrix of $F(y)$. As shown in Figure 4, $N \times 128$ local features from the source and the target point cloud are fed into the feature embedding and fusion layer. Matrices $M(x, y)$, $M(x, x)$ and $M(y, y)$ containing the transformation information T^1 between $F(x)$ and $F(y)$, are obtained using the transformations given in Equations (9), (10) and (12).

Moreover, in order to fuse more features, the max-pooling function is used to extract the global feature f_x of the source point cloud and the global feature f_y of the target point cloud, respectively. Then similarly we calculate the inner product matrix $M(f_x, f_y)$ between f_x and f_y .

Therefore, T^1 can be computed using these four matrices, thereby the T can be achieved from T^1 . However the inverse matrices of $M(x, y)$ and $M(x, x)$ may not exist, thus we perform feature extraction directly on these four matrices. A 2D convolutional network is used to fuse the features in matrices $M(x, y)$, $M(x, x)$, $M(y, y)$ and $M(f_x, f_y)$. The feature embedding and fusion layer is composed of stacked convolution layers. In the convolution layers, the input is down-sampled with a step size of 2, instead of the pooling. All the layers, except the last one, are followed by batches normalization, and the ReLU activation function is used. The 2D CNNs settings are Conv2d (4, 128, 128)—Conv2d (8, 64, 64)—Conv2d (16, 32, 32)—Conv2d (8, 16, 16). The convolution kernel has a size of 3×3 . We also expand the last layer of 2DCNN into a one-dimensional tensor of 2048 as an input to the next Parametric regression layer.

4.4. Parametric regression

The fusion features are provided as input to the multiple fully connected layers that compute the estimated transformation T . As shown in Figure 4, the parametric regression layer has three hidden layers of size 1024, 512 and 256, respectively, and an output layer of size 8. The output of size 8 is $[q_0, q_1, q_2, q_3, x_t, y_t, z_t, s_{est}]^T$. The first four elements of this vector are the rotation quaternion $q \in \mathbb{R}^4$, the middle three forms translation vectors $t_{est} \in \mathbb{R}^3$ and the scale factor $s_{est} \in \mathbb{R}^1$. Four tuples are used to transform the rotation matrix. In addition, q is converted into $R_{est} \in SO(3)$ as

$$[q_0, q_1, q_2, q_3] = \frac{[q_0, q_1, q_2, q_3]}{\sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}} \quad (13)$$

$$R = \begin{pmatrix} 1 - 2q_2^2 - 2q_3^2 & 2q_1q_2 - 2q_0q_3 & 2q_1q_3 + 2q_0q_2 \\ 2q_1q_2 + 2q_0q_3 & 1 - 2q_1^2 - 2q_3^2 & 2q_2q_3 - 2q_0q_1 \\ 2q_1q_3 - 2q_0q_2 & 2q_2q_3 + 2q_0q_1 & 1 - 2q_1^2 - 2q_2^2 \end{pmatrix} \quad (14)$$

Therefore, the predicted target point cloud is given by

$$Y_{est} = s_{est} * X * R_{est} + t_{est} \quad (15)$$

4.5. Loss function

The loss function used to train the registration network aims at minimizing the distance between the corresponding points on the source and target point clouds. The EMD is an efficient function that is

used in PCRNet [SLG*19] to measure the distance between the predicted target point cloud Y_{est} and the target point cloud Y :

$$Loss_{EMD(Y_{est}, Y)} = \min_{Y_{est} \rightarrow Y} \frac{1}{N} \sum_{Y_{est} \in Y} \|Y - Y_{est}\|_2 \quad (16)$$

where N denotes the number of points in the point cloud, Y represents the target point cloud and Y_{est} is the predicted target point cloud transformed by using Equation (15). However, when there are problems such as partial overlap and multi-scale between the source and target point clouds. It will fall into a local optimum. Therefore, the EMD function is used between the predicted target point cloud Y_{est} and the corresponding target point cloud Y_{gt} :

$$Y_{gt} = s \otimes X \otimes R + t \quad (17)$$

where $(s, R, t) \in T$, T denotes the ground truth.

$$Loss_{EMD(Y_{est}, Y)_{gt}} = \min_{Y_{est} \rightarrow Y_{gt}} \frac{1}{N} \sum_{Y_{est} \in Y} \|Y_{gt} - Y_{est}\|_2 \quad (18)$$

Figure 7 illustrates the relationship between Y , Y_{gt} and Y_{est} . It is important to mention that Y_{gt} is obtained from the source point cloud through ground truth transformation, which is not the original given target point cloud Y . Therefore, in the process of reverse propagation, outliers can be effectively identified and given a smaller weight, thus the registration of partial-partial overlap point clouds can be handled. In contrast to the direct use of ground truth, our loss function does not need to take into account the weights between the different metrics. The Adam optimizer is used to optimize our network.

5. Results

5.1. ModelNet40 dataset

The experiment was conducted on the ModelNet40 dataset [WSK*15], which contains CAD models from 40 artificial object categories. For each given shape of the dataset, we follow PointNet, and a random sampling of 2048 points is used to form a point cloud and unitise it into a sphere. ModelNet40 includes 12,311 grid CAD models, of which 80% are used as training set and 20% are used as testing set. For each point cloud, we randomly select 1024 points as the source point cloud.

As for the ground truth T , we randomly sample three Euler angles with the rotation range of $[0, 90]$ to generate the rotation matrix, three numbers in the range of $[-1, 1]$ to generate translation vectors and a number in the range of $[0.2, 0.8]$ to generate the scale factor. Note that the rotation range in all the references [LZX*19, FLLW21] is all set to $[0, 45]$, whereas it is set to $[0, 90]$ in our work. The transformed point cloud is regarded as the target point cloud.

In order to generate partially overlapping point clouds, we adopt the same method as PrNet [WS19b], since it is closer to the real-world point clouds, that is, fixes a randomly point far away from two point clouds and sample 768 points closest to the fixed point in two point clouds. For more details, refer to PrNet [WS19b].

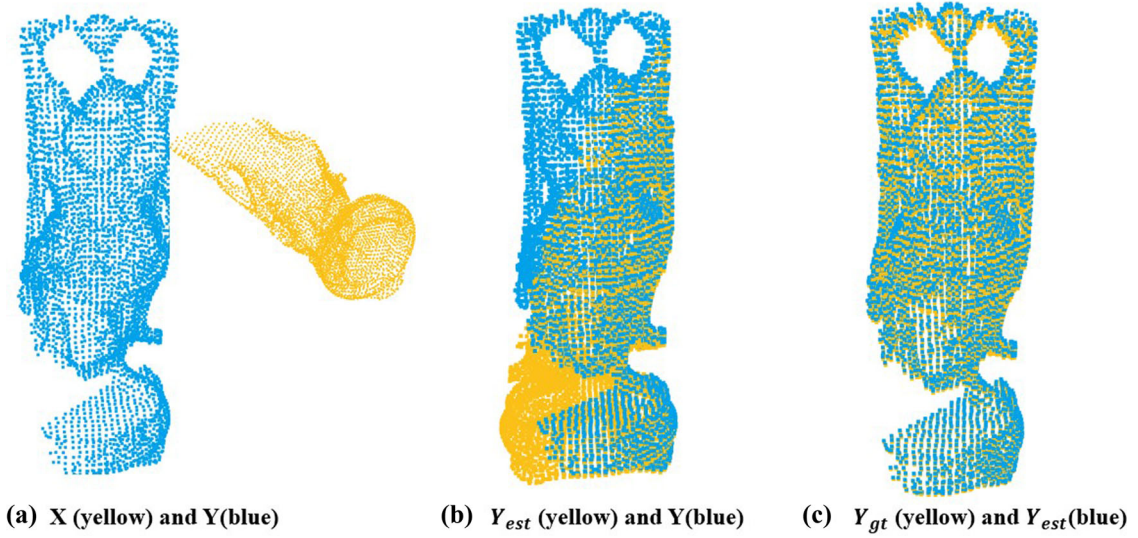


Figure 7: As can be seen from (b), there will be a large number of false matches if we employ the EMD function between the predicted target point cloud Y_{est} and target point cloud Y .

5.2. Baseline algorithm and evaluation metrics

To make a comprehensive comparison, seven typical methods are selected as benchmarks. The first three methods (ICP [BM92], FGR [ZPK16], and SICP [ZXH*17]) are traditional classical algorithms, while the other four methods (PCNet [SLG*19], IDAM [LZX*19], RGMNet [FLLW21] and GMCNet [PCL21]) are recent representative deep learning-based approaches. We use the ICP and FGR implemented in Intel Open3D to make a fair and comprehensive comparison. Since the code of SICP is not provided, we implemented it according to the published paper, while all its parameters are finetuned. The codes of PCNet, IDAM, RGMNet and GMCNet were implemented according to the one provided by the authors on GitHub. They were also trained in our dataset. Five evaluation metrics are used: the mean isotropic errors (MIE) of rotation and translation proposed by RPMNet [YL20], the mean absolute errors (MAE) of Euler angle and translation vector proposed by DCP [WS19a], that are anisotropic, as well as the recall rate. Recall rate for all our experiments was set as $MIE(R) < 15$ and $MIE(t) < 0.3$.

5.3. Partial overlap

Depending on the scanning method, most of the scanned point clouds are not complete, resulting in a partial-to-partial point cloud registration. In fact, this is the most challenging situation. The points having no corresponding points are considered as outliers [FLLW21]. Therefore, we first test the registration performance on partially overlapping point clouds. The details of generating partially overlapping point clouds are presented in Section 5.1. Note that, although the number of points is equal, a one-to-one correspondence between the source and the target point clouds does not necessarily exist, and all the point clouds in the training sets and testing sets are partially overlap.

Table 1 shows the results obtained by the proposed method and the selected baseline algorithms. It can be seen that for a point

Table 1: Performance on partially overlap data.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
ICP	80.56	0.42	40.42	0.21	25.00%
PCR	20.19	0.15	16.99	0.07	64.26%
FGR	16.71	0.10	10.29	0.05	81.40%
SICP	102.97	0.52	57.88	0.25	26.94%
IDAM	11.25	0.04	6.99	0.02	86.60%
RGM	8.52	0.06	7.32	0.03	91.10%
GMC	0.41	0.02	0.39	0.02	100.0%
Ours	3.76	0.04	3.42	0.02	94.32%

cloud registration problem with only partial overlap challenges, the results of GMCNet are optimal and our model closely follows. However, our model takes only one-ninth of the time needed by GMCNet. Moreover, GMCNet is exceptionally sensitive to noise and multi-scale, while our model is more robust to noise and multi-scale. The local registration methods ICP and SCIP produce the worst performance because of the large initial rotation angle between the point clouds. Furthermore, the performance of PCNet, FGR and RGMNet cannot compete with that of HDRNet. Some qualitative comparisons are shown in Figures 8a and 9.

5.4. Gaussian noise

In this section, we evaluate the robustness of our model to Gaussian noises, which usually exist in real world point clouds. Similar to Li *et al.* [LZX*19], we first randomly generate a series of numbers of standard deviation of 0.01, and the size is clipped at $[-0.05, 0.05]$ then add them to the point cloud as Gaussian noise. The experiments in Section 5.3 are repeated.

Table 2 shows the results obtained by the HDRNet and the selected baseline algorithms, on these noisy data. It can be seen that

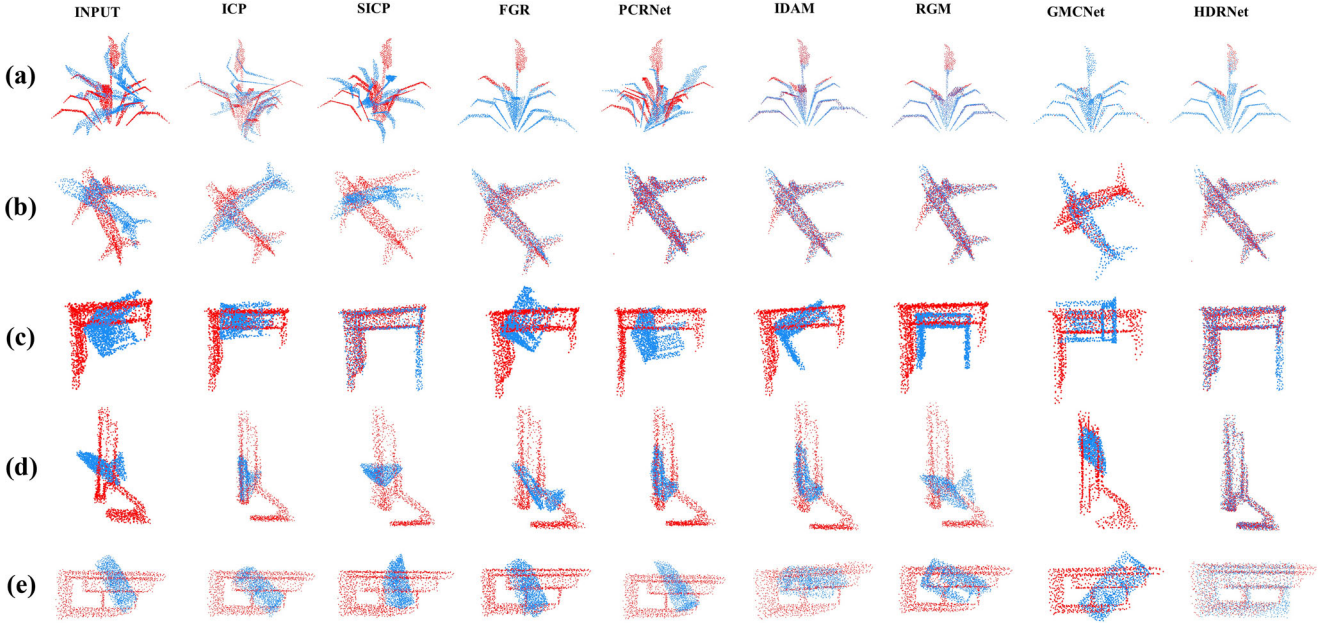


Figure 8: Qualitative registration results on ModelNet40, (a) partial overlap, (b) noise, (c) and (d) different scales, and (e) unseen categories.

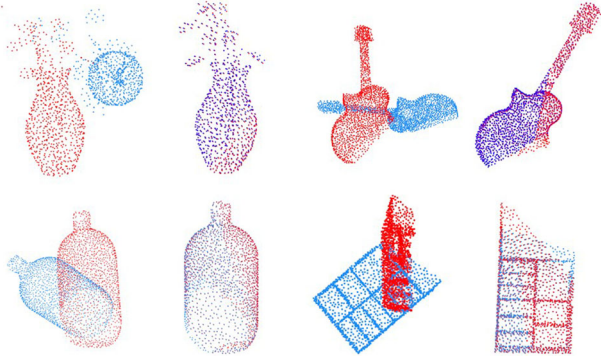


Figure 9: Example results of partial overlap on ModelNet40 dataset.

Table 2: Performance on point clouds with Gaussian noise.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
ICP	79.88	0.42	40.07	0.20	28.72%
PCRnet	16.98	0.14	12.11	0.06	64.10%
FGR	68.51	0.40	40.20	0.19	28.36%
SICP	99.40	0.50	55.79	0.24	28.84%
IDAM	32.68	0.14	21.10	0.07	58.20%
RGM	12.23	0.10	9.80	0.04	83.36%
GMC	59.67	0.21	36.75	0.24	26.35%
Ours	3.97	0.04	3.65	0.02	94.06%

the proposed method has a higher success rate than other methods. More precisely, the obtained success rate is very close to that obtained in the case where the Gaussian noise does not exist. This

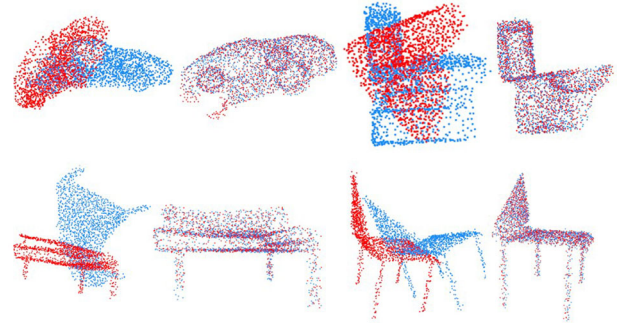


Figure 10: Example results of partial overlap and noise on ModelNet40 dataset.

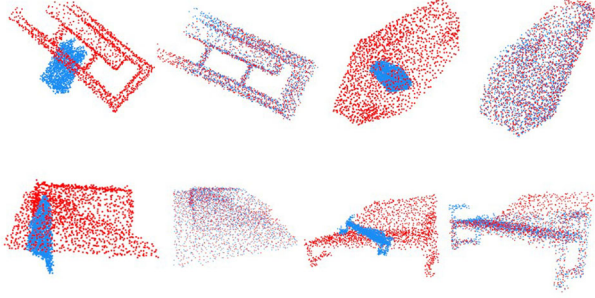
demonstrates that the proposed model is robust to Gaussian noise. However, FGR and GMCNet are very sensitive to noise, so the success rate of registration is only 28.36% and 26.35%. IDAM and RGMNet are also susceptible to Gaussian noise. On the contrary, ICP, SICP and PCRNet are more robust to Gaussian noise. However, their performance is still not satisfactory. The proposed method outperforms all the learning and handmade methods. Some qualitative comparisons are shown in Figures 8b and 10.

5.5. Multi-scales

In this section, we evaluate the performance of the HDRNet on point clouds with multi-scales. In order to generate point clouds of multi-scales, we randomly generate a number in the range of [0.2, 0.8] and only apply it to the source point cloud, resulting in two point clouds exhibiting the same shape, but in multi-scales.

Table 3: Performance on multi-scale data.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
ICP	88.13	0.50	44.00	0.25	9.44%
PCR	18.95	0.18	16.05	0.08	61.30%
FGR	87.50	0.51	45.73	0.25	5.22%
SICP	98.62	0.48	54.78	0.23	28.76%
IDAM	28.48	0.21	15.81	0.10	40.72%
RGM	47.11	0.29	28.51	0.15	35.21%
GMC	68.64	0.22	42.30	0.25	20.66%
Ours	4.34	0.05	4.17	0.03	92.87%

**Figure 11:** Example results of partially overlapping, noise, and multi-scales on ModelNet40 dataset.

Afterwards, the sampling process described in Section 5.3 is repeated to obtain the partially overlapped point clouds. Furthermore, the method used to generate noise in Section 5.4 is also repeated to create noisy point clouds.

All the models are retrained on multi-scale datasets. Table 3 presents the obtained results. Since the scale factor is introduced, it can be seen that the proposed model can effectively deal with the point clouds having diverse scales. Furthermore, the proposed method highly outperforms all the selected baseline methods, with a success rate on multi-scale data sets close to the case where noise is used alone. Since RGMNet, IDAM, ICP, PCRNet and GMCNet do not consider the scale information, they cannot deal with this case, and the results obtained by SICP are still affected by the initial large rotation angles. The proposed method outperforms all the learning and handmade methods. Some qualitative comparisons are shown in Figures 8c,d and 11.

5.6. Invisible categories

This section deals with the evaluation of the HDRNet performance on the invisible category of point clouds. In order to generate different types of point clouds, the ModelNet40 dataset is splitted. The first 30 categories are chosen to generate, the training set and test on the remaining 10 categories. Afterwards, we retrain our model, PCRNet, IDAM, RGMNet and GMCNet on the training set, then evaluate them on the testing set. ICP, SICP and FGR are also tested on the same testing set. The obtained results are summarized in Table 4. The detailed experimental results reveal that our method still achieves the highest success rate. It can also be seen that the se-

Table 4: Performance on invisible category point clouds.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
ICP	87.68	0.52	44.46	0.26	8.66%
PCR	22.29	0.20	17.98	0.10	47.00%
FGR	88.31	0.52	45.50	0.26	5.16%
SICP	102.51	0.51	58.85	0.25	27.66%
IDAM	30.83	0.20	18.90	0.10	33.16%
RGM	54.01	0.35	31.71	0.17	22.83%
GMC	72.43	0.22	48.76	0.25	17.21%
Ours	7.07	0.06	5.93	0.03	87.16%

Table 5: Performance on stanford 3D scanning repository.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
ICP	111.75	0.51	61.70	0.27	10%
PCR	20.45	0.17	18.06	0.09	50%
FGR	108.79	0.63	57.02	0.40	10%
SICP	30.64	0.23	23.75	0.12	30%
IDAM	28.15	0.21	17.53	0.15	30%
RGM	53.51	0.42	26.59	0.20	20%
GMC	72.72	0.22	46.20	0.23	20%
Ours	5.19	0.07	8.83	0.03	90%

lected learning-based methods perform worse if they are not trained on the last 10 categories. The proposed method outperforms all the learning and handmade methods. Some qualitative comparisons are shown in Figure 8e.

5.7. Evaluation on real data

To evaluate the generality of our model on real data, we further tested our model on the Stanford 3D Scanning repository. This dataset has 10 scans, and for each scan, we generated the target point cloud by randomly transforming the scan in the same way as in Section 5.1. We also used the approach in PRNet to generate partially overlapping point clouds. The trained model on the ModelNet40 dataset is directly employed for registration. Table 5 shows the results obtained by HDRNet and the selected baseline algorithm on the Stanford 3D Scanning repository. The experiment was evaluated in the multi-scale setting. It can be seen that our method still has a higher success rate than the other baseline methods. Some qualitative examples are shown in Figure 12.

5.8. Ablation studies

In this section, we conducted ablation experiments to demonstrate the role of each component in the architecture. All experiments are evaluated on a multi-scale setting. We analysed the following two key components:

To demonstrate the effectiveness of the corresponding matrix module, we directly used the high-dimensional features output by the local feature extraction module without transformation by the corresponding matrix estimation module, and the resulting method

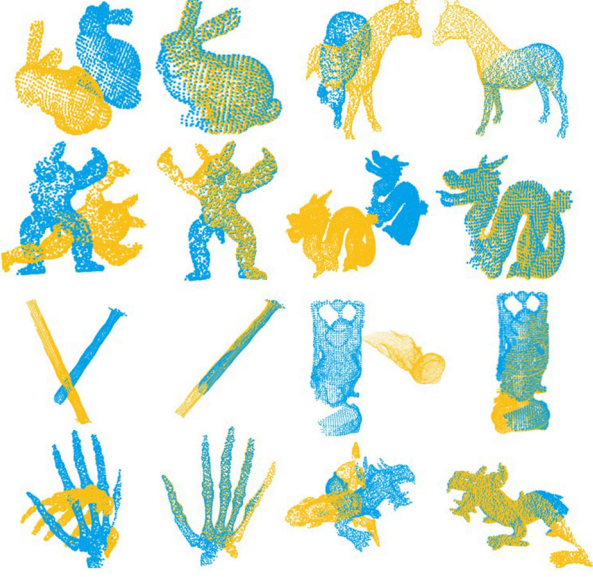


Figure 12: Example results on Stanford 3D scan.

Table 6: Ablation studies—key components.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
HDRVar1	6.03	0.06	5.47	0.04	88.76%
HDRVar2	14.99	0.11	14.51	0.09	67.45%
HDRVar3	8.62	0.05	6.48	0.09	82.37%
HDRVar4	42.83	0.30	26.24	0.14	25.86%
HDRNet	4.34	0.05	4.17	0.03	92.87%

is denoted as HDRVar1. The results are listed in the first row of Table 6. We found that the registration error was significantly reduced after using the correspondence matrix module.

To demonstrate the importance of the feature embedding and fusion modules, we designed a variant that is directly concatenated using the global features of the source and target point clouds, and denote the resulting method as HDRVar2. The result is displayed in the second row of the Table 6. It is evident that the registration accuracy becomes very poor after using the variant of the feature embedding and fusion module, which indicates that the proposed feature embedding and fusion module can effectively improve the registration performance.

To demonstrate the effectiveness of the scale factor, we removed the scale factor from the end of the model and obtained a network called HDRVar3. The results are listed in the third row of Table 6. We find that the scale factors are effective for multi-scale datasets.

To demonstrate the advantage of using only the global position of a single point to establish point correspondence, we use the Euclidean distance in 3D space directly without using Equation (1) and represent the resulting method as HDRVar4. The results are shown in the fourth row of Table 6. It is clear that the alignment accuracy becomes poor after using the Euclidean distance in 3D space, indi-

Table 7: Ablation studies—loss function.

Network	loss function	MIE(R)	MIE(t)	Recall
HDRNet	EMD(Y_{est}, Y)	9.43	0.08	81.50%
	EMD(Y_{est}, Y_{gt})	4.34	0.05	92.87%
PCRNet	EMD(Y_{est}, Y)	18.95	0.18	61.30%
	EMD(Y_{est}, Y_{gt})	14.41	0.13	73.34%

Table 8: Ablation studies—loss function.

Method	MIE(R)	MIE(t)	MAE(R)	MAE(t)	Recall
GTPC	5.29	0.06	6.65	0.05	88.76%
EMD	4.34	0.05	4.17	0.03	92.87%

Table 9: The average time (ms) required for registration of point clouds with different number of points.

# Points	512	1024	2048
ICP	4.14	9.16	18.43
PCR	30.79	34.84	36.01
FGR	11.40	18.13	33.90
SICP	440.07	1251	3462
IDAM	32.28	32.92	35.51
RGM	105.34	214.75	724.87
GMC	91.14	108.02	153.78
Ours	16.29	16.96	17.46

cating that using the global position of a single point can be effective in improving the alignment performance.

To demonstrate the efficiency of the proposed $EMD(Y_{est}, Y_{gt})$ loss function, the architecture performance is also tested by replacing the $EMD(Y_{est}, Y)$ loss function in PCRNet. The obtained results are shown in Table 7. Note that experiments are also carried out on HDRNet and PCRNet. It can be observed comparing lines 1-2 and 3-4, that using $EMD(Y_{est}, Y_{gt})$ loss function can effectively improve the success rate.

Empirically, the use of EMD gives better results than the known ground truth point correspondences (denoted as GTPC) to calculating the loss, the experimental results are shown in Table 8.

5.9. Efficiency

In this section, the computational efficiency of the proposed network is tested and compared with that of ICP, PCRNet, SICP, FGR, IDAM, RGMNet and GMCNet. The speed on point clouds is tested with 512, 1024 and 2048 points, respectively.

We ran the experiment on a 3.59GHz AMD Ryzen7 3700 CPU and an Nvidia GeForce RTX2080TI GPU. For all the models, we experiment with a batch of one size. ICP, SICP and FGR ran on CPU, while the other algorithms including the proposed algorithm, run on GPU. We averaged the time of all the models on the testing set. The detailed information is provided in Table 9.

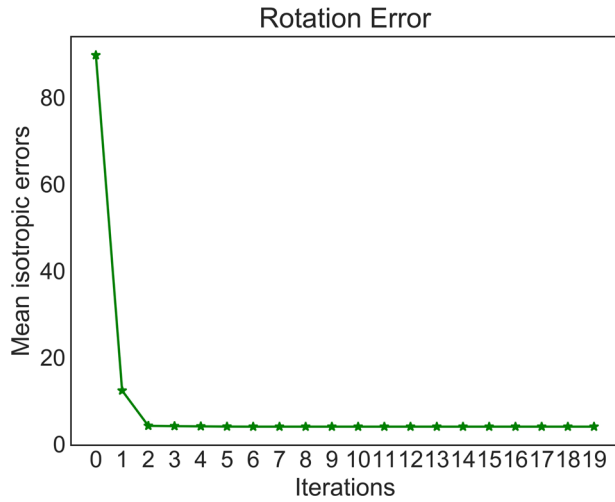


Figure 13: The mean isotropic errors over registration iterations.

It can be seen from Table 9 that the proposed method is the fastest compared with the three learning-based methods. It can also be noticed that, when the number of points is 2048, our method is optimal. However, in the other two cases, it is slower than ICP.

How many iterations are needed? Figure 13 shows the MIE after each iteration. Most of the performance improvements occurred in the first two iterations, and the registration converged after the first two iterations (which we use for all experiments). This contributes to the high efficiency of the proposed network.

6. Conclusion

This paper proposed a novel end-to-end point cloud registration network, which directly consumes point clouds. Instead of using max-pooling to derive the global characteristics in previous methods, our method applies the Gaussian function to orthogonalize the corresponding matrix, which is used to extract the corresponding relationship between the source and target point clouds. This highly promotes the availability of local point cloud features, and can effectively deal with the point cloud registration with large initial angles. Furthermore, a scale factor capable of handling multi-scale point cloud registration is introduced. Owing to the improved loss function, the proposed network can effectively handle the registration of partially overlapping point clouds. Compared with the traditional methods and the recently proposed learning-based methods of the literature, the obtained experimental results highlight the efficiency of the proposed network.

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